Mir Sazzat Hossain

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Education

Independent University, Bangladesh **Jan 2017 – Apr 2021**
Bachelor of Science in ***Computer Science and Engineering*** *Dhaka, Bangladesh*
Minor in ***Engineering Mathematics***
CS & Math GPA: ***3.81/4.0*** *(Overall CGPA:* ***3.64/4.0****)*

Work Experience

AI Engineer **Aug 2025 – Present**
Penta Global Limited (Part-time Aug 2025 – Jul 2026, Full-time since Aug 2026) *Dhaka, Bangladesh*

- Developing **Plinth**, an agentic AI platform that turns 2D architectural and structural drawing PDFs into complete 3D BIM models in Autodesk Revit.
- Designing a **LangGraph**-based multi-agent pipeline where vision-language models read drawings and extract building components, and agents build the full model in Revit through **MCP**.
- Built a chat assistant that lets users interact with BIM models in Autodesk Revit (part-time role).

Research Experience

Post-baccalaureate Research Assistant **Feb 2022 – July 2026**
Center for Computational & Data Sciences, Independent University, Bangladesh *Dhaka, Bangladesh*

- **Research Interests:** LLM Agents, Test Time Reinforcement Learning (TTRL), On-Policy Self-Distillation, Adversarial Policy Optimization, Vision-Language Models, Formal Verification & RLVR.
- **Teaching Support:** Introduction to Programming, Data Structures, AI, ML, and Image Processing.
- **Publications:** Q1 Journals ×3, ACL (A*), WACV (A), ICIP ×3 (B), AACL (B), IJCNN (B), and multiple workshop papers, including one in AAAI.

Publications

(C) Rafid Ahmed, Intesar Tahmid, **Mir Sazzat Hossain**, Tasnimul Hossain Tomal, Md Mahir Jawad, Anam Borhan Uddin, Md Fahim, and Md Farhad Alam Bhuiyan. "**Evaluating Large Vision Language Models on Bangla Medical Visual Question Answering**" In Findings of the Association for Computational Linguistics: ACL 2026, pages 37362–37378, San Diego, California, United States. Association for Computational Linguistics. **[CORE Rank: A*]** [Paper]

(C) Md Fahim, Md Farhan Ishmam, **Mir Sazzat Hossain**, Md. Ashraful Amin, Amin Ahsan Ali, and A K M Mahbubur Rahman. "**R-MMA: Enhancing Vision-Language Models with Recurrent Adapters for Few-Shot and Cross-Domain Generalization**" in Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2026, pp. 6496-6506. **[CORE Rank: A]** [Paper]

(C) **Mir Sazzat Hossain**, A K M Mahbubur Rahman, Md. Ashraful Amin and Amin Ahsan Ali. "**Lightweight Recurrent Neural Network for Image Super-Resolution**" 2024 IEEE International Conference on Image Processing (ICIP), Abu Dhabi, United Arab Emirates, 2024, pp. 1567–1573, DOI: 10.1109/ICIP51287.2024.10647844. **[CORE Rank: B]** [Paper] [Code]

(C) Rakibul Hasan Rajib, Md Akil Raihan Iftee, **Mir Sazzat Hossain**, A K M Mahbubur Rahman, Sajib Mistry, Md. Ashraful Amin and Amin Ahsan Ali. "**FedCTTA: A Collaborative Approach to Continual Test-Time Adaptation in Federated Learning**" 2025 International Joint Conference on Neural Networks (IJCNN), Rome, Italy, 2025, pp. 1-8, DOI: 10.1109/IJCNN64981.2025.11228519. **[CORE Rank: B]** [Paper] [Code]

(C) Newaz Ben Alam, Akm Moshiur Rahman Mazumder, **Mir Sazzat Hossain**, Mysha Samiha, Md Alvi Noor Hossain, Md Fahim, Amin Ahsan Ali, Ashraful Islam, Md. Ashraful Amin, and A K M Mahbubur Rahman. "**CMBan: Cartoon-Driven Meme Contextual Classification Dataset for Bangla.**" In Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics, pages 2178–2194, Mumbai, India. The Asian Federation of Natural Language Processing and The Association for Computational Linguistics. **[CORE Rank: B]** [Paper]

(C) **Mir Sazzat Hossain**, Ovi Paul, Md Akil Raihan Iftee, Rakibul Hasan Rajib, Abu Bakar Siddik Nayem, Anis Sarker, Arshad Momen, Md. Ashraful Amin, Amin Ahsan Ali, and A K M Mahbubur Rahman. "**BD Open LULC Map: High-Resolution Land Use Land Cover Mapping & Benchmarking for Urban Development in Dhaka, Bangladesh**" 2025 IEEE International Conference on Image Processing (ICIP), Anchorage, Alaska, USA, 2025, pp. 2808–2813, DOI: 10.1109/ICIP55913.2025.11084396. **[CORE Rank: B]** [Paper]

(C) **Mir Sazzat Hossain**, K M B Asad, Payaswini Saikia, Adrita Khan, Md Akil Raihan Iftee, Rakibul Hasan Rajib, Arshad Momen, Md. Ashraful Amin, Amin Ahsan Ali, and A K M Mahbubur Rahman. "**RGC-Bent: A Novel Dataset for Bent Radio Galaxy Classification**" 2025 IEEE International Conference on Image Processing (ICIP), Anchorage, Alaska, USA, 2025, pp. 2868–2873, DOI: 10.1109/ICIP55913.2025.11084387. **[CORE Rank: B]** [Paper] [[Code]]

(J) **Mir Sazzat Hossain**, Md Shahadat Hossain Shahal, K M B Asad, Payaswini Saikia, Adrita Khan, Fatema Akter, Amin Ahsan Ali, Md. Ashraful Amin, Deba Priyo Guha, Mahady Hasan, M. O. B. Jihad, Arshad Momen, Siddhartho Sen, and A K M Mahbubur Rahman. "**RGC: A Radio AGN Classifier Based on Deep Learning. I. A Semi-Supervised Multiclass Model for VLA Images**" in Astronomy & Astrophysics (A&A), vol. 714, pp. A5, 2026, DOI: 10.1051/0004-6361/202557828. **[Q1 Journal]** [Paper] [Code]

(J) Fahim Mahafuz Ruhad, Md Fahim, **Mir Sazzat Hossain**, Md Fahad Monir, Ashraful Islam, and Md. Ashraful Amin, "**Beyond Classification: Benchmarking Object Detection Models for Efficient Tomato Leaf Disease Identification on a Real-World Dataset**" in Elsevier Smart Agricultural Technology, vol. 12, pp. 101336, 2025, DOI: 10.1016/j.atech.2025.101336. **[Q1 Journal]** [Paper]

(J) Team BD Lensing. "**Investigating the Relation Between Environment and Internal Structure of Massive Elliptical Galaxies Using Strong Lensing**" in Astronomy & Astrophysics (A&A), vol. 699, pp. A259, 2025, DOI: 10.1051/0004-6361/202453239. **[Q1 Journal]** [Paper] [Project]

(W) Rafid Ahmed, Intesar Tahmid, **Mir Sazzat Hossain**, Tasnimul Hossain Tomal, Md Fahim, Md Farhad Alam Bhuiyan. "**How Good LLMs Are at Answering Bangla Medical Visual Questions? Dataset and Benchmarking**" Proceedings of The Second AAAI Bridge Program on AI for Medicine and Healthcare, PMLR 317:1-14, 2026. [Paper]

(W) Md Mahir Jawad, Rafid Ahmed, Ishita Sur Apan, Tasnimul Hossain Tomal, Fabiha Haider, **Mir Sazzat Hossain**, and Md Farhad Alam Bhuiyan. "**Benchmarking Large Language Models on Bangla Dialect Translation and Dialectal Sentiment Analysis**". In Proceedings of the Second Workshop on Bangla Language Processing (BLP-2025), pages 322–337, Mumbai, India. Association for Computational Linguistics. [**Received Best Long Paper Award**] [Paper] [Award]

(W) Tasnimul Hossain Tomal, Anam Borhan Uddin, Intesar Tahmid, **Mir Sazzat Hossain**, Md Fahim, and Md Farhad Alam Bhuiyan. "**LP-FT-LoRA: A Three-Stage PEFT Framework for Efficient Domain Adaptation in Bangla NLP Tasks**". In Proceedings of the Second Workshop on Bangla Language Processing (BLP-2025), pages 212–222, Mumbai, India. Association for Computational Linguistics. [Paper]

(W) **Mir Sazzat Hossain**, Sugandha Roy, K M B Asad, Arshad Momen, Amin Ahsan Ali, Md. Ashraful Amin and A K M Mahbubur Rahman. "**Morphological Classification of Radio Galaxies Using Semi-Supervised Group Equivariant CNNs**" in Procedia Computer Science, vol. 222, pp. 601–612, 2023, ISSN: 1877-0509, DOI: 10.1016/j.procs.2023.08.198. [Paper] [Code]

Teaching Experience

Post-baccalaureate Teaching Assistant **May 2021 – Jan 2022**

Independent University, Bangladesh, Dept. of Computer Science and Engineering *Dhaka, Bangladesh*

- Prepared the lab materials, provided programming support, and addressed student queries during lab sessions for Numerical Methods, Neural Networks, and Data Mining courses.

Undergraduate Teaching Assistant **Oct 2019 – Apr 2021**

Independent University, Bangladesh, Department of Physical Sciences *Dhaka, Bangladesh*

- Offered one-on-one tutoring to struggling students from various disciplines at Math Tutorial Center.
- Worked with other tutors to manage the center and make sure students got the support they needed.

Awards, Honors & Achievements

- Received the **Best Long Paper Award** - BLP Workshop @ IJCNLP–AACL 2025. [Credentials]
- Received the **IEEE Signal Processing Society Travel Grant** of **US $1,000** for joining the **2024 IEEE ICIP** in Abu Dhabi, UAE (October 27-30). [Credentials]
- IUB academic honors: **VC's List (3×)**, **Dean's Merit List**, and **Dean's List**. [Credentials]
- **Winner:** Intra IUB Tech Fest Programming Contest (Summer 2019) and ACM Week Code Debugging Contest (2019). [Tech Fest] [ACM Week]
- Represented IUB in the **ICPC Asia Dhaka Regional 2019** and participated in various workshops, programming contests, and Olympiads. [ICPC] [Extracurricular Activities]

Relevant Coursework

Artificial Intelligence, Machine Learning, Image Processing, Numerical Methods, Linear Algebra and Differential Equation, Multivariable Calculus, Ordinary Differential Equation, Boundary Value Problem

Standardized Tests

- **IELTS**: Overall 7.0 (Reading - 8.0, Listening - 7.5, Speaking - 7.0, Writing - 6.0)

Technical Skills

- **Languages**: Python, C++, C, JavaScript
- **Libraries**: PyTorch, PyTorch Lightning, LLaMA-Factory, Hugging Face, Flower, Scikit-learn
- **Web Frameworks**: Next.js, React, Django, Laravel
- **Miscellaneous**: Git, Docker, Apptainer, HTCondor, Slurm, MATLAB, LaTeX

References

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